Fetal Health Multiclassification Using a Reduced Feature Set

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1. Problem Statement

Cardiotocography (CTG) records fetal heart rate and uterine contractions during pregnancy. The task is to classify each CTG record into one of three fetal-state classes: Normal, Suspect or Pathological. Our assigned reference, a Stanford CS229 report, selects 10 of the 21 CTG features with SelectKBest and reaches 95.58% test accuracy. Our goal was to rebuild the same pipeline with only **5 features**, so that a model needs half the measurements, and to see how much accuracy that costs.

2. Dataset

We use the UCI Cardiotocography dataset (ID 193): 2,126 CTG records with 21 numeric features and the 3-class NSP label assigned by consensus of three expert obstetricians. The data contains **13 exact duplicate records**. We remove them, leaving 2,113 records (1,646 Normal, 292 Suspect, 175 Pathological), the same as the reference paper. Without this step a record could appear in both the training and the test set, which inflates test accuracy. The classes are imbalanced (78% Normal), so we report macro and per-class F1 alongside accuracy.

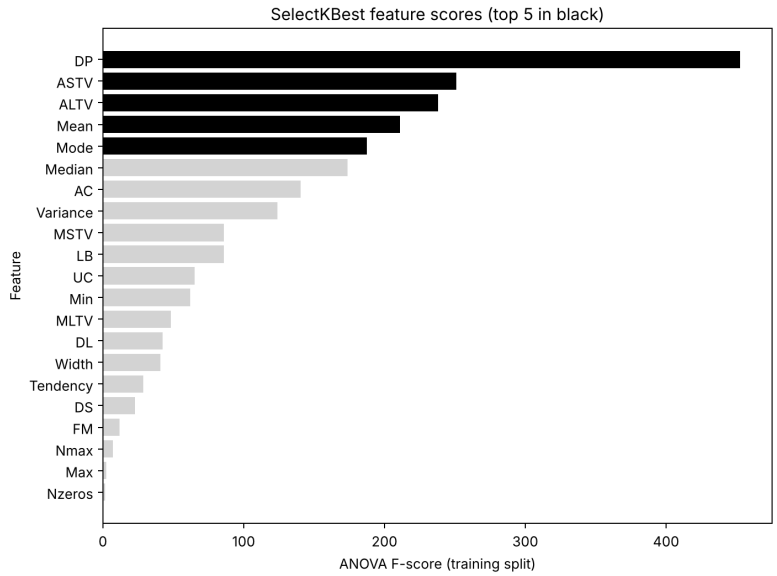
3. Approach

We follow the reference paper's pipeline: (1) remove duplicates; (2) stratified 70/30 train-test split (seed 42; 1,479 train, 634 test); (3) StandardScaler; (4) SelectKBest with the ANOVA F-test, keeping k = 5 features; (5) train ten classifiers. The scaler and selector are fitted on the training split only, so the test set plays no part in choosing features. Each model is lightly tuned with a small grid search using 5-fold stratified cross-validation on the training split, and evaluated once on the test set. The five features selected are: **DP, ASTV, ALTV, Mean, Mode** (prolonged decelerations, abnormal short-term variability, percentage of time with abnormal long-term variability, and the histogram mode and mean).

We also considered PCA. With the same split and an untuned Random Forest, 5 PCA components give 89.75% accuracy against 93.53% for the 5 SelectKBest features. PCA builds components that keep the most variance, not the most class information, and it mixes features together, which removes the simple single-feature thresholds that tree models rely on. SelectKBest was therefore kept.

4. Implementation

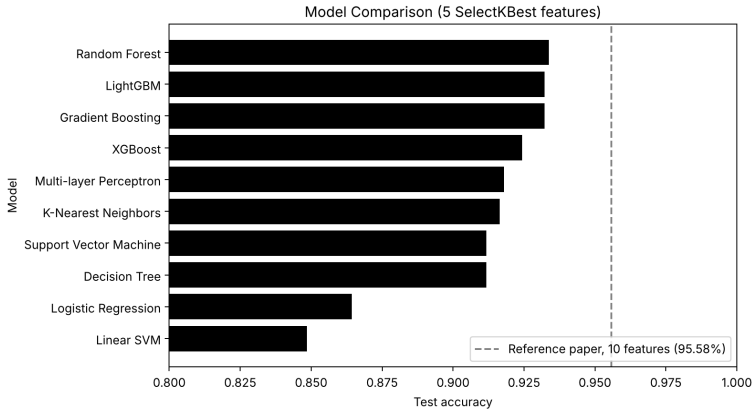
The project is written in Python with pandas, scikit-learn, XGBoost and LightGBM. `download_dataset.py` builds the dataset, `preprocessing.py` handles duplicates, the split, scaling and SelectKBest, `models.py` defines the ten models and their grids, `evaluation.py` computes metrics and confusion matrices, and `run_experiments.py` runs the full pipeline and writes every table and figure in `results/`. A Jupyter notebook walks through the same steps for the live demonstration.

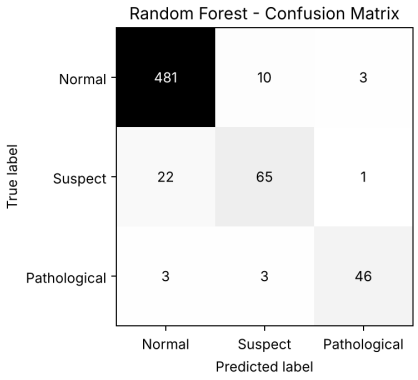


5. Results

The best model is **Random Forest** with **93.38%** test accuracy (592/634 records) and macro F1 0.882. Gradient Boosting and LightGBM follow closely. Tree ensembles clearly outperform the linear models, which cannot separate the Suspect class well using only five features.

Model	Accuracy	Macro F1	Normal F1	Suspect F1	Path. F1	CV acc.
Random Forest	**0.9338**	**0.882**	**0.962**	**0.783**	**0.902**	**0.9317**
Gradient Boosting	0.9322	0.887	0.959	0.788	0.913	0.9256
LightGBM	0.9322	0.886	0.960	0.766	0.932	0.9277
XGBoost	0.9243	0.876	0.954	0.763	0.913	0.9304
Multi-layer Perceptron	0.9180	0.846	0.955	0.736	0.846	0.9196
K-Nearest Neighbors	0.9164	0.841	0.956	0.729	0.837	0.9121
Decision Tree	0.9117	0.847	0.947	0.722	0.872	0.9128
Support Vector Machine	0.9117	0.830	0.952	0.688	0.851	0.9121
Logistic Regression	0.8644	0.722	0.930	0.519	0.716	0.8742
Linear SVM	0.8486	0.671	0.924	0.467	0.622	0.8587





6. Conclusions

Using only 5 of the 21 CTG features, our best model reaches 93.38% accuracy. That is 2.2 percentage points below the reference paper's 95.58%, which uses 10 features, so the 5-feature model does not beat the paper. In exchange, it needs half the inputs. Normal and Pathological records are still recognised well (F1 about 0.96 and 0.90). Most of the loss is in the Suspect class (F1 0.78), which sits between the other two and depends on features that were dropped, such as accelerations and short-term variability. Removing the 13 duplicate records makes the evaluation honest, and fitting the scaler and selector on the training data only prevents information leaking from the test set. SelectKBest outperformed PCA at the same size. A natural next step is to choose features with a model-based or class-weighted criterion aimed at the Suspect class.

Limitations: results come from a single 70/30 split of a public benchmark, so a difference of one or two test records (0.16% each) between models is within noise. This is an academic prototype, not a clinically validated diagnostic system.

Data: UCI Cardiotocography, DOI 10.24432/C51S4N. Reference: Ereyi and Huo, Multiclass Classification of Fetal Health using Cardiotocogram Data, Stanford CS229, Spring 2021. All numbers above, apart from the paper's 95.58%, come from our own runs.